

Notes: Futia (QJE, 1980)

Schumpeterian Competition

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1 Big picture

- **Question.** What kind of dynamic industry model can explain the observed connection between market structure and technological innovation?
- **Setting.** The paper is a purely theoretical study of a single industry. Industry evolution is modeled as a stochastic process of innovation, imitation, and entry, with firms repeatedly choosing R&D effort.
- **Core challenge.** Empirically, concentration and innovation are related in complicated ways. More concentrated industries often spend more on R&D, but this relationship weakens when innovative opportunities are abundant, and innovative effort may peak at intermediate concentration levels. A useful theory must explain these patterns jointly.
- **Main conceptual contribution.** Futia builds a distinctly Schumpeterian model in which firms innovate because successful innovation can *change market structure* and temporarily create market power. But those rents are later eroded by imitation, entry, and future innovation.
- **Main empirical regularities the theory targets.**
 - R&D input is positively correlated with concentration.
 - That positive correlation is weaker in industries with greater innovative opportunity.
 - Industries with greater innovative opportunity tend to be more concentrated.
 - R&D input relative to sales may be highest at intermediate, not extreme, concentration levels.
- **Main modeling strategy.** The industry state is (n, j) : n firms coexist in short-run equilibrium and all use technique or product j . Innovation moves the technology frontier from j to $j + 1$. Entry and imitation determine how many firms survive and follow the innovation.
- **Main equilibrium objects.** The model delivers a stationary distribution over industry sizes. From that distribution the paper defines:
 - long-run expected number of firms, n^* ,
 - long-run expected innovation probability, μ^* .
- **Main baseline result.** Lower entry barriers reduce concentration in the long run, but they also weaken each firm's incentive to innovate because successful innovation becomes less valuable when rents are quickly dissipated.
- **Key mechanism result.** For a given industry size, the innovation probability rises with the number of firms, because innovation is more attractive when many rivals can be displaced. But easier entry lowers the gain from winning the innovation race, so its overall effect on long-run innovation is ambiguous.
- **Key theoretical conclusion.** On average, the model predicts a positive relationship between concentration and innovation, but it can also generate an inverted-U between them when the direct negative effect of easier entry on innovation is eventually offset by the positive effect operating through more firms.

- **Main credibility result.** The model delivers these predictions with explicit choice-theoretic foundations: innovation probabilities come from a one-period R&D game, and long-run results come from the stationary distribution of the induced Markov chain.
- **Economic takeaway.** The Schumpeterian link between market power and innovation does not come from current size per se. It comes from the prospect that innovation can temporarily reshape industry structure and create rents before rivals imitate, enter, and innovate again.

2 Model environment and key objects

2.1 Empirical regularities the paper is designed to explain

- The paper does not estimate the model directly. Instead, it asks whether a coherent dynamic theory can account for four stylized facts from the empirical R&D literature.
- Those facts concern the relationship among concentration, innovative opportunity, and aggregate R&D effort relative to sales.

Interpretation. The model is judged by whether its comparative statics line up with broad qualitative evidence, not by a formal structural estimation exercise.

2.2 Industry state: short-run equilibrium (n, j)

A short-run industry equilibrium is summarized by a pair of integers

$$(n, j), \tag{1}$$

where:

- n is the number of identical firms coexisting in the market,
- j indexes the currently prevailing product or production technique.

The interpretation of j is deliberately flexible:

- in a process-innovation reading, higher j means lower cost,
- in a product-innovation reading, higher j means higher quality.

The ordering is decisive: technique or product j dominates k whenever $k < j$.

Interpretation. The model collapses all within-industry heterogeneity into two state variables: how many firms survive, and how advanced the common technology is.

2.3 Innovation process and sure survivors

Let N_t and J_t denote industry size and technology at date t . The paper assumes

$$\Pr(J_{t+1} = J_t + 1 \mid J_t = j, N_t = n) = \mu(n), \quad (2)$$

$$\Pr(J_{t+1} = J_t \mid J_t = j, N_t = n) = 1 - \mu(n). \quad (3)$$

Thus innovation occurs one step at a time and, crucially, the probability depends only on industry size n , not on the current technology level j .

To track who certainly survives the innovation race, Futia defines S_t , the number of “sure survivors.” If no firm innovates, all n firms survive for sure; if innovation occurs, only the single innovator is guaranteed to survive.

$$\Pr(S_t = n \mid N_t = n, J_t = j) = 1 - \mu(n), \quad (4)$$

$$\Pr(S_t = 1 \mid N_t = n, J_t = j) = \mu(n). \quad (5)$$

Interpretation. Innovation is modeled as a winner-take-the-lead event. But the innovator’s advantage is temporary because imitation and entry can still replenish the market in the next period.

2.4 Entry condition v and the imitation kernel

The paper introduces an exogenous “condition of entry,” denoted $v \in [0, 1]$:

- low v : entry is difficult,
- high v : entry is easy.

Given $S_t = l$, the next period number of firms is random with conditional distribution

$$\lambda(l, k, v) = \Pr(N_{t+1} = k \mid S_t = l). \quad (6)$$

The key assumptions are:

- **A1.** $\sum_{k=1}^{\infty} \lambda(l, k, v) = 1$, and λ is continuous in v .
- **A2.** $\lambda(l, k, v) = 0$ if $k < l$.
- **A3.** There is an upper bound $\bar{n}$ on the number of firms that can coexist.
- **A4.** For any fixed m , $\sum_{k=1}^m \lambda(l, k, v)$ decreases with l and with v .
- **A5.** If $v = 0$, then $\lambda(l, l, 0) = 1$; if $v = 1$, then $\lambda(l, \bar{n}, 1) = 1$.

Interpretation.

- v captures how quickly innovation rents are competed away.
- A4 says that more sure survivors or easier entry shifts the next-period size distribution to the right.

- A5 pins down the extremes: when entry is blocked, no new imitators arrive; when entry is free, the industry jumps to the competitive upper bound $\bar{n}$.

2.5 Long-run equilibrium distribution

Given $\mu(n)$ and $\lambda(l, k, v)$, the transition probability for industry size alone is

$$p(n, m) = \Pr(N_{t+1} = m \mid N_t = n) = \mu(n)\lambda(1, m, v) + [1 - \mu(n)]\lambda(n, m, v). \quad (7)$$

The paper then studies the Markov chain $\{N_t\}$.

Interpretation. The technology index J_t keeps rising over time, so there is no stationary distribution over (N_t, J_t) . But because innovation probabilities do not depend on j , the industry-size process $\{N_t\}$ can still have a stationary distribution.

2.6 Two summary statistics: concentration and pace of innovation

If $\pi^*(n)$ is the stationary probability of observing n firms, the paper defines

$$n^* = \sum_{n=1}^{\bar{n}} n\pi^*(n), \quad (8)$$

$$\mu^* = \sum_{n=1}^{\bar{n}} \mu(n)\pi^*(n). \quad (9)$$

Here:

- n^* is the long-run expected number of firms,
- μ^* is the long-run expected innovation probability.

Because more firms mean a less concentrated industry, concentration is inversely related to n^* .

Interpretation. These are the paper's central endogenous objects: long-run market structure and long-run innovative pace.

2.7 Rents and innovative opportunity

At short-run equilibrium (n, j) , each of the n firms earns rents of the form

$$\theta(n)R(j). \quad (10)$$

The paper assumes:

- $\theta(1) = 1$, $\theta(n) > 0$,
- $n\theta(n)$ is nonincreasing in n ,
- $\theta(n)$ is decreasing and convex in n ,

- $R(j) = \gamma^j R$ for some $\gamma \geq 1$.

A useful example is

$$\theta(n) = n^{-\beta}, \quad \beta \geq 1. \quad (11)$$

Interpretation.

- $\theta(n)$ governs how fast rents erode as rivalry increases.
- γ indexes “innovative opportunity”: higher γ means a successful innovation creates a larger proportional jump in monopoly rents.

2.8 Major simplifications of the setup

The model is intentionally stylized:

- only one technology or product quality survives at a time,
- innovation happens one step at a time,
- only firms already inside the industry perform relevant R&D,
- entry occurs through imitation, not independent outside innovation,
- firm size gives no direct cost advantage in R&D.

Interpretation. The paper sacrifices realism to keep the dynamic logic tractable and to isolate the Schumpeterian idea that innovation matters because it temporarily rearranges market power.

3 Models and theoretical specifications (all equations + interpretation)

3.1 Markov transition structure of the industry

The full state process has transition probabilities

$$\Pr(N_{t+1} = m, J_{t+1} = j \mid N_t = n, J_t = i) = \begin{cases} 0, & j \neq i, i+1, \\ \mu(n)\lambda(1, m, v), & j = i+1, \\ [1 - \mu(n)]\lambda(n, m, v), & j = i. \end{cases} \quad (12)$$

Interpretation.

- If innovation occurs, technology improves by one step and the next-period number of firms is drawn from the one-survivor distribution.
- If innovation does not occur, technology stays the same and the next-period number of firms is drawn from the n -survivor distribution.

3.2 Theorem A: existence of long-run stochastic equilibrium

Under A1 and the condition $\inf_{l \geq 2} \mu(l) > 0$, Theorem A states that there exists a unique stationary distribution π^* over industry sizes such that

$$\sum_{n=1}^{\bar{n}} \pi^*(n) p^t(n, m) = \pi^*(m) \quad (13)$$

for all m and t , and convergence to π^* occurs from any initial distribution at a geometric rate.

Interpretation.

- The industry never settles at one fixed size forever.
- But it does settle into a stable long-run probability law over sizes.
- This makes it meaningful to talk about average concentration and average innovative pace.

3.3 One-period R&D game

At state (n, j) , each firm chooses an R&D input $x_i \in [0, 1]$. The objective is to maximize expected next-period rents net of current R&D cost.

The paper assumes:

$$p(x_1, \dots, x_n) = a \sum_{i=1}^n x_i, \quad (14)$$

where p is the probability that at least one firm innovates, and

$$\alpha_i(x_1, \dots, x_n) = \frac{x_i}{\sum_{k=1}^n x_k}, \quad (15)$$

where α_i is firm i 's probability of winning the race conditional on an innovation occurring.

R&D cost is assumed quadratic:

$$C(x_i) = \frac{b}{2} x_i^2. \quad (16)$$

The cost is scaled by γ , so both benefits and costs grow with innovative opportunity.

Interpretation. A higher x_i makes innovation more likely and increases the firm's chance of being the winner, but at increasing marginal cost.

3.4 Expected rents and the approximation used by the paper

Let

$$M(l, v) = \sum_{k=1}^{\bar{n}} k \lambda(l, k, v) = E[N_{t+1} \mid S_t = l]. \quad (17)$$

The paper also makes a simplifying assumption that firms approximate the expected rent term

$$\eta(l, v) = \sum_{k=1}^{\bar{n}} \lambda(l, k, v) \theta(k) \quad (18)$$

by

$$\theta(M(l, v)). \quad (19)$$

Interpretation. Firms evaluate the effect of innovation and non-innovation through the expected number of firms next period, rather than through the full distribution of next-period firm counts.

3.5 Dominant-strategy equilibrium R&D input

Under the linear innovation technology, the proportional win rule, and quadratic costs, each firm's optimal choice is independent of rivals' choices. The dominant-strategy input level is

$$x^*(n, v, \gamma) = \frac{a}{b} R [\gamma \theta(M(1, v)) - \theta(M(n, v))] . \quad (20)$$

Interpretation.

- A firm invests more in R&D when winning is much better than the no-innovation outcome.
- The gain from winning is larger when one innovator can hold the market down to few firms, and smaller when entry quickly expands the market back toward many firms.

3.6 Innovation probability induced by equilibrium behavior

The resulting innovation probability is

$$\mu(n, v, \gamma) = a n x^*(n, v, \gamma). \quad (21)$$

This completes the model: the innovation process is no longer primitive but is derived from optimizing behavior.

Interpretation. Long-run innovation is jointly determined by the number of firms, the profitability of temporary market power, and how easy it is for entry and imitation to erode that power.

3.7 Why the short-run innovation probability rises with the number of firms

From the expression for $x^*(n, v, \gamma)$, an increase in n raises $M(n, v)$, lowers $\theta(M(n, v))$, and therefore raises the gain from winning. Because

$$\mu(n, v, \gamma) = a n x^*(n, v, \gamma), \quad (22)$$

there are also simply more firms trying to innovate.

Interpretation. For a *fixed* entry environment, less concentrated industries can have higher innovation probabilities because more firms are competing for a prize that can radically change market structure.

3.8 Why easier entry lowers the incentive to innovate for a fixed n

The paper adds assumption A12: $D_v M(l, v)$ is nonincreasing in l . Combined with the convexity of θ , this implies that

$$\frac{\partial \mu(n, v, \gamma)}{\partial v} < 0 \quad (23)$$

for each fixed n .

Interpretation. Easier entry means the winner of the innovation race will face more imitators and new entrants, so the reward from winning shrinks. That weakens R&D incentives holding industry size fixed.

3.9 Determinants of long-run concentration

The paper distinguishes short-run from long-run comparative statics.

Short run.

- Higher v makes entry easier and pushes the next-period size distribution toward more firms.
- Lower γ reduces R&D incentives, which means innovation happens less often and all firms are more likely to survive, again pushing toward more firms.

Long run. To ensure that these short-run effects are not overturned by feedback from future innovations, Futia imposes assumption A13, a stochastic monotonicity condition ensuring that the expected next-period number of firms rises with the current number of firms.

Under A13:

- increasing v increases n^* ,
- decreasing γ increases n^* .

Because concentration is inversely related to n^* , this implies:

- easier entry lowers long-run concentration,
- greater innovative opportunity raises long-run concentration.

Interpretation. Strong innovative opportunities make market power more valuable, so the long-run industry tends to spend more time in concentrated states.

3.10 Long-run concentration and innovation

The paper next studies $\mu^*(v)$, the long-run pace of innovation as a function of the entry condition.

First, $\mu^*(v)$ is continuous in v because the state space is finite and the stationary distribution varies continuously with the transition matrix.

Second, the paper shows that, in general,

$$\mu^*(0) > \mu^*(1). \quad (24)$$

The logic comes from the polar cases:

- if $v = 0$, entry is blocked and the stationary distribution collapses to monopoly, $n = 1$;
- if $v = 1$, entry is free and the stationary distribution collapses to the competitive upper bound, $n = \bar{n}$.

Then, using the equilibrium innovation formula, the payoff from successful innovation is larger under monopoly than under free entry, because aggregate rents fall with the number of firms.

Interpretation. This gives the model's average positive relationship between concentration and innovation: more concentrated industries innovate more than very competitive ones, because innovation creates larger temporary rents there.

3.11 How the model generates an inverted-U

Although $\mu^*(0) > \mu^*(1)$, the paper argues that innovation need not be highest under complete entry blockade. The curve $\mu^*(v)$ can peak at an intermediate v .

Because a closed-form derivative is intractable, Futia introduces a deterministic “certainty-equivalent” analogue:

$$\hat{n}_{t+1} = \mu(\hat{n}_t, v, \gamma)M(1, v) + [1 - \mu(\hat{n}_t, v, \gamma)]M(\hat{n}_t, v), \quad (25)$$

with stable fixed point $\hat{n}$, and defines

$$\hat{\mu}(v) = \mu(\hat{n}(v, \gamma), v, \gamma). \quad (26)$$

The sign of $D_v \hat{\mu}(v)|_{v=0}$ is determined by two opposing forces:

- a **direct negative effect**: easier entry lowers the reward from winning,
- an **indirect positive effect**: easier entry raises the number of firms, and more firms raise the innovation probability.

If industry rents do not fall too rapidly as concentration decreases, the indirect effect can dominate near monopoly, so the innovation curve peaks at an intermediate entry condition.

Interpretation. This is the paper's explanation for why innovation can be highest at moderate concentration rather than at monopoly or perfect competition.

4 What makes the results go through (assumptions, logic, and limitations)

4.1 Why v is treated as exogenous

The model is about how legal or technological entry conditions shape innovation and market structure. Futia explicitly notes that some real-world barriers to entry, such as advertising or limit pricing, are themselves chosen by firms. Those are excluded from v .

Interpretation. The comparative statics are clean only because v is taken as outside the industry's control.

4.2 Why successful innovation raises concentration in the short run

Under A4, the expected number of firms next period is smaller when there is one sure survivor than when all current firms survive:

$$E[N_{t+1} \mid S_t = 1] < E[N_{t+1} \mid S_t = n]. \quad (27)$$

Interpretation. Innovation temporarily concentrates the market, because the winner can eliminate current rivals before imitation replenishes the industry.

4.3 Why easier entry has an ambiguous effect on long-run innovation

Two effects work in opposite directions:

- easier entry weakens the value of innovation,
- easier entry creates less concentrated industries, and those have more firms trying to innovate.

Interpretation. This is the heart of the model. Entry policy changes both the *prize* for innovation and the *number of contestants*. That is why a simple reduced-form concentration-innovation regression can be misleading.

4.4 Role of the stochastic monotonicity assumption A13

A13 rules out a “cobweb” possibility in which easier entry could initially produce more firms, which then causes enough extra innovation to reduce the number of firms in the long run. The condition ensures that larger current industries translate into stochastically larger future industries.

Interpretation. A13 is not a trivial assumption; it is what lets short-run direction map into long-run direction for concentration.

4.5 What is proved exactly and what is heuristic

- The existence and uniqueness of the stationary distribution are proved using Markov-chain arguments.
- The sign results for concentration under A13 are also formal.
- The inverted-U result for innovation is developed heuristically through the certainty-equivalent model and then checked numerically in the appendix.

Interpretation. The strongest result in the paper is the existence of a coherent stochastic equilibrium with sensible comparative statics. The midpoint maximum for innovation is persuasive but not fully proved in closed form.

4.6 What the model leaves out

The paper intentionally abstracts from several realistic features:

- coexistence of multiple technologies or qualities,
- outside innovators,
- perfect patent protection,
- dynamic strategic behavior beyond one-period R&D choice,
- direct scale advantages in R&D.

Interpretation. The model is a theory of Schumpeterian competition through temporary rents, not a full theory of industrial innovation.

4.7 Relation to the literature and to Schumpeter

The model sits between two ideas in the older literature:

- the Schumpeterian claim that market power and innovation are linked,
- the empirical finding that the relation is neither mechanically monotone nor stable across industries.

Futia's contribution is to show that both can be true once market structure itself is endogenous and continuously reshaped by innovation and entry.

5 Key theoretical results

5.1 The four empirical regularities in the introduction

Read:

- The introduction states four stylized facts: more concentration is associated with more R&D, this positive relation weakens when innovative opportunity is high, high-opportunity industries are more concentrated, and innovation may peak at moderate concentration.
- These are not estimated inside the paper; they are the scorecard for the theory.

Economic message. The paper is a theory-building exercise aimed at jointly rationalizing several empirical regularities that otherwise look hard to reconcile.

5.2 Theorem A: stationary distribution over industry size

Read:

- Theorem A establishes a unique stationary distribution π^* over the number of firms.
- Convergence to that distribution occurs from any initial condition.

Economic message. This is what makes the model operational. Without π^* , there would be no coherent way to define long-run concentration or long-run innovation in a stochastic Schumpeterian world.

5.3 Equation (1) for equilibrium R&D input

Read:

$$x^*(n, v, \gamma) = \frac{a}{b} R [\gamma \theta(M(1, v)) - \theta(M(n, v))] . \quad (28)$$

This is the core choice equation in the paper.

Economic message.

- R&D is high when winning the innovation race sharply improves the firm's post-innovation rent position.
- Entry conditions matter because they determine how strongly innovation reshapes market structure.

5.4 Section IV: long-run concentration

Read:

- Under stochastic monotonicity, increasing entry ease v raises n^* , meaning the market becomes less concentrated.
- Increasing innovative opportunity γ lowers n^* , meaning the market becomes more concentrated.

Economic message. Strong innovative opportunity does not simply raise innovation. It also tilts the long-run market structure toward more concentrated states because the rents from market leadership become more valuable.

TABLE I
CORRELATION BETWEEN μ^* AND $(-n^*)$
 $v = 0.1, 0.2, 0.3, 0.4, 0.5, 0.6$

β	γ		
	1.1	1.5	2.0
1	0.668	0.486	0.174
2	0.900	0.886	0.873

5.5 Section V: average positive relation and possible inverted-U

Read:

- The paper shows that generally $\mu^*(0) > \mu^*(1)$: monopoly with blockaded entry innovates more than free entry, because innovation creates bigger rents there.
- But the certainty-equivalent argument shows that $\mu^*(v)$ can peak at an interior value of v .

Economic message. The model can simultaneously deliver Schumpeter's positive concentration-innovation intuition and the empirical inverted-U pattern.

5.6 Appendix Table 1: the two-state numerical example

Read:

- The appendix studies a simple case with only two industry sizes, $n = 1$ and $n = 10$.
- It sets $\lambda(1, 1, v) = 1 - v$, $\lambda(1, 10, v) = v$, and $\lambda(10, 10, v) = 1$, so $M(1, v) = 1 + 9v$ and $M(10, v) = 10$.
- With $\theta(n) = n^{-\beta}$, $a = 0.1$, $b = 1$, and $R = 5$, the numerical results confirm the heuristic predictions.
- For fixed v , larger innovative opportunity γ raises μ^* .
- The positive correlation between innovation and concentration weakens as γ rises. Table 1 reports, for example, that when $\beta = 1$, the correlation between μ^* and $(-n^*)$ falls from 0.668 at $\gamma = 1.1$ to 0.174 at $\gamma = 2.0$.

Economic message. The numerical example is small, but it is important because it shows the paper's central comparative statics are not just verbal. The simple stochastic model actually behaves the way the intuition says it should.

6 Short summary

- Futia builds a stochastic Schumpeterian industry model in which firms innovate to gain temporary market power, but imitation and entry later dissipate those rents.
- Industry evolution is summarized by a Markov process over the number of firms and the current technology level.
- A one-period R&D game endogenizes the innovation probability.

- Long-run concentration and long-run innovation are defined from the stationary distribution over industry size.
- Easier entry lowers concentration but also weakens the incentive to innovate, so its net effect on long-run innovation is ambiguous.
- On average, the model predicts more innovation in more concentrated industries, yet it can also generate an inverted-U with innovation peaking at intermediate concentration.
- Greater innovative opportunity raises concentration and weakens the positive concentration-innovation correlation, matching the broad empirical literature.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that a dynamic Schumpeterian model with optimizing R&D, temporary innovation rents, imitation, and entry can jointly determine market structure and innovation in the long run. The theory is rich enough to explain several seemingly separate empirical patterns with one coherent mechanism.

7.2 Economic intuition

The key idea is simple: innovation matters because it can temporarily reduce rivalry and create market power. But the size of that prize depends on how easy it is for imitation and entry to undo the innovator's advantage. Policy or technology that affects entry therefore changes both the number of firms and the reward to innovation.

7.3 Takeaway

Futia's main message is that one cannot study concentration and innovation in a static framework where market structure is fixed. Innovation both responds to and reshapes market structure, and the long-run relationship between them depends on the tension between temporary monopoly rents and the speed with which competition returns.